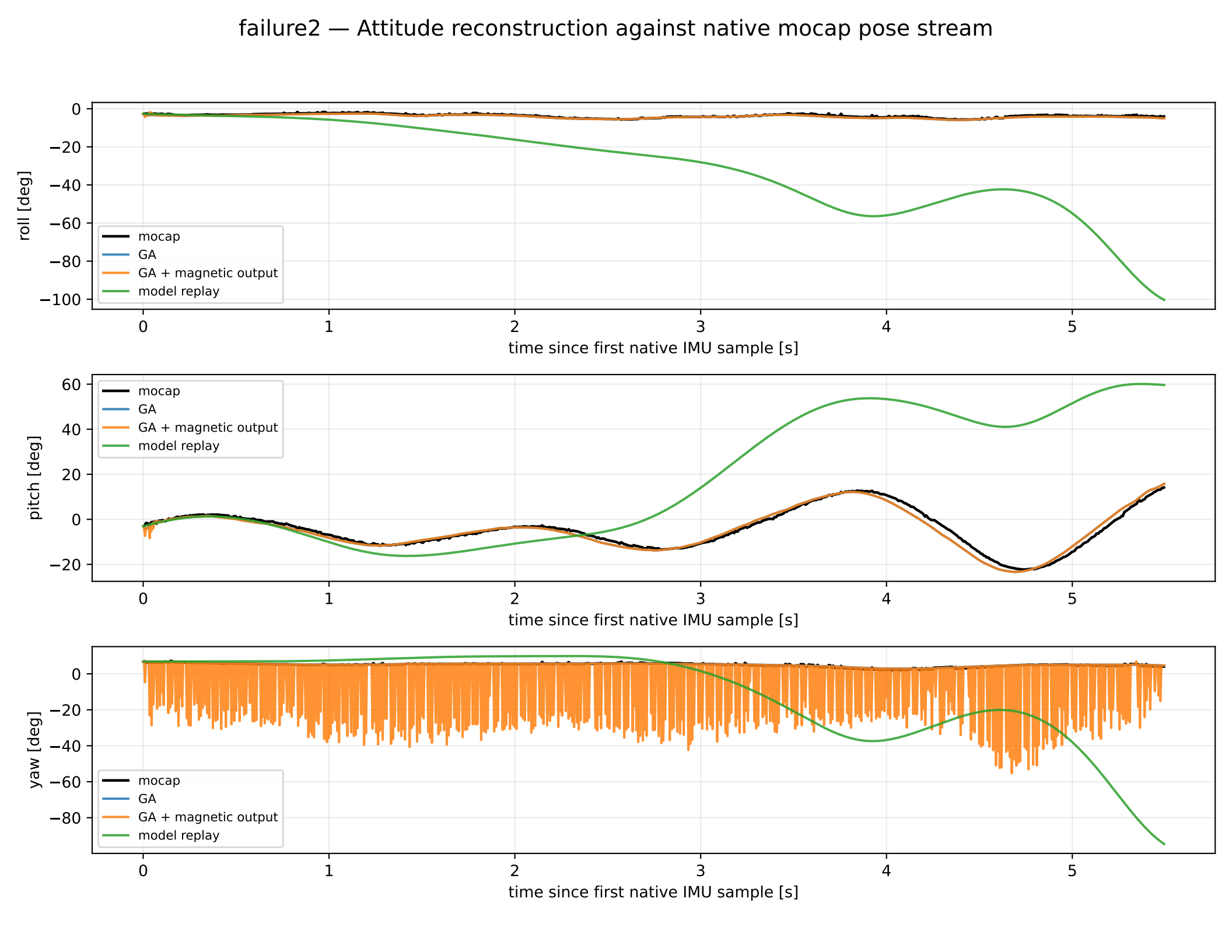
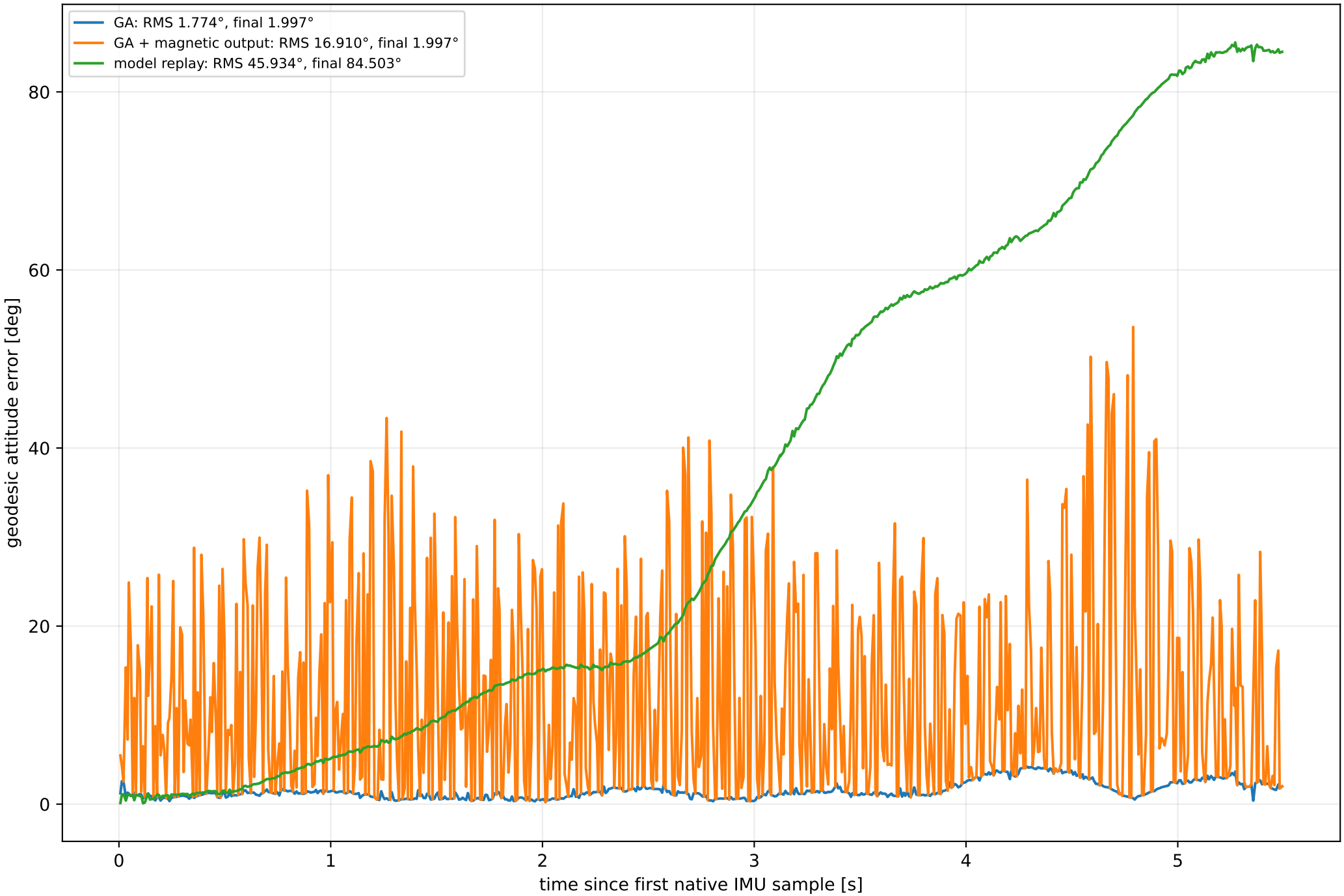
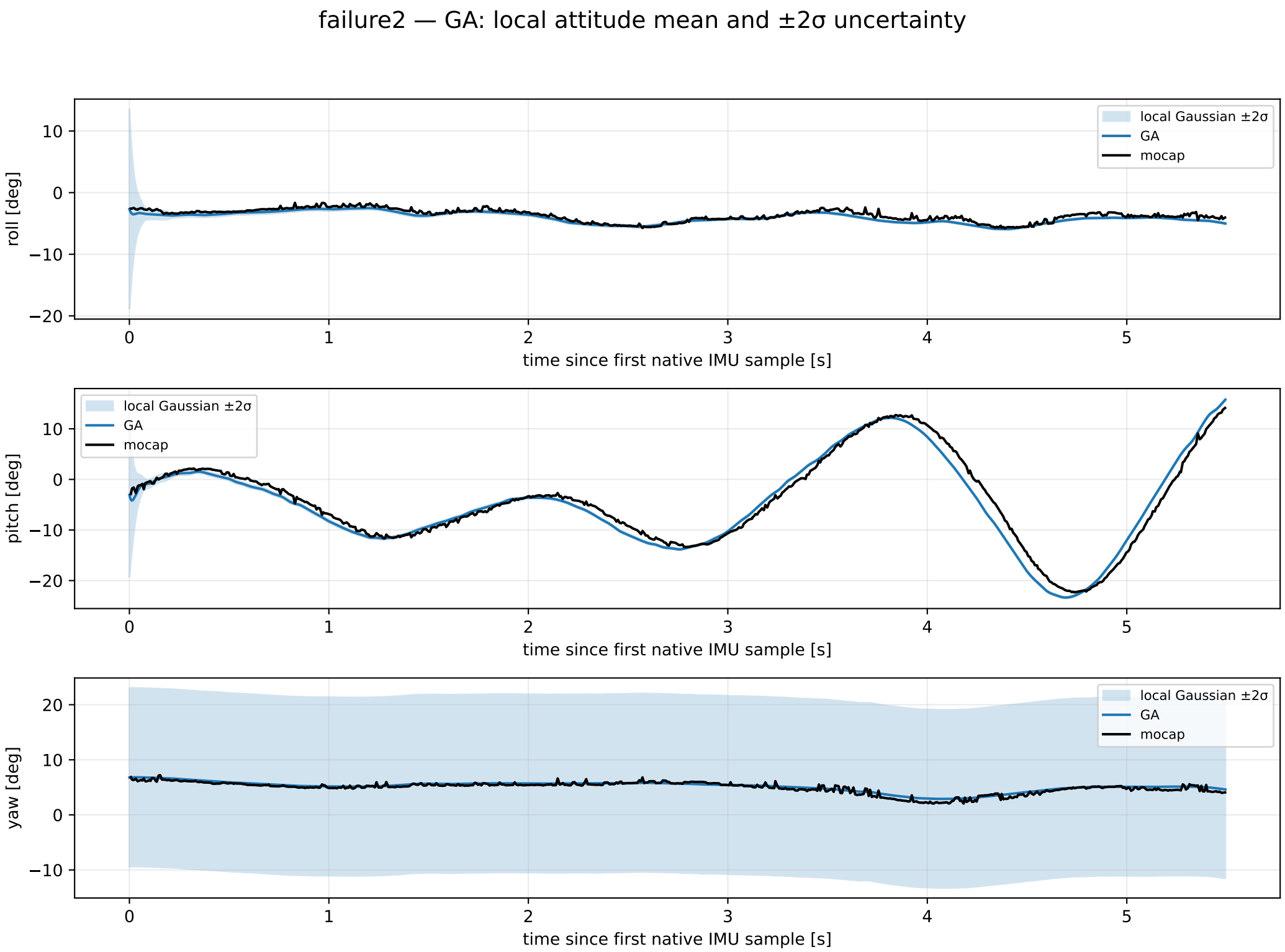
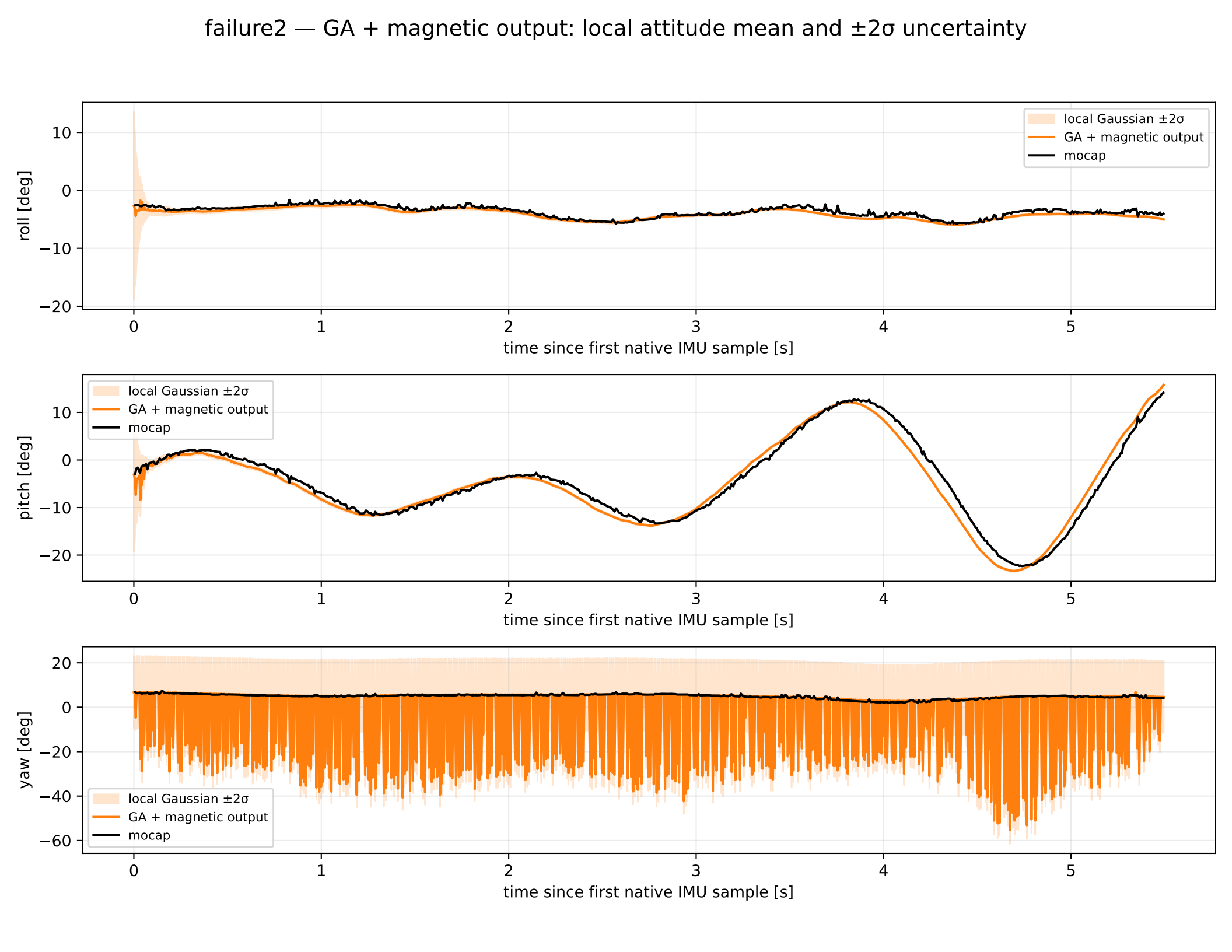
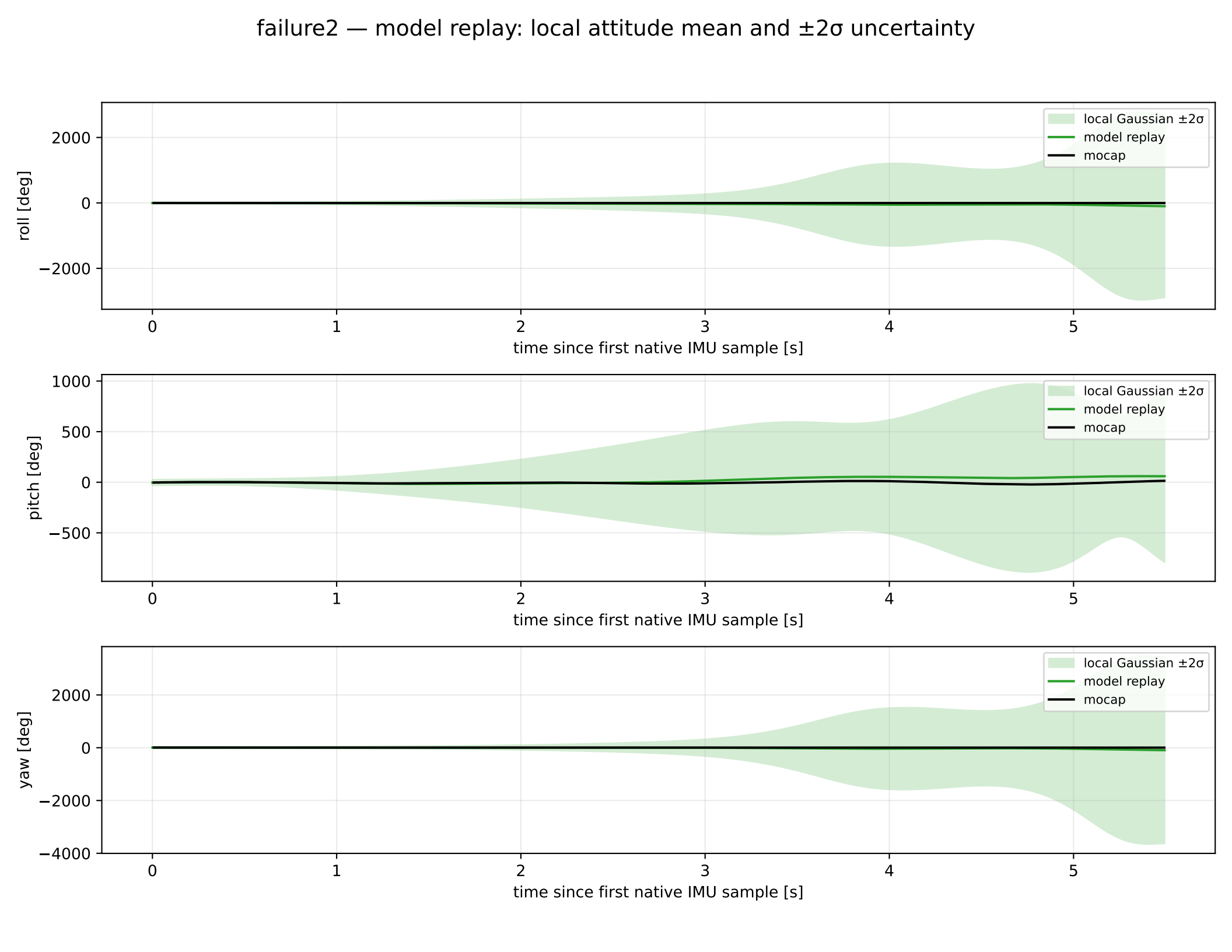

failure2 — Attitude reconstruction against native mocap pose stream
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
mocap
GA
GA + magnetic output
model replay

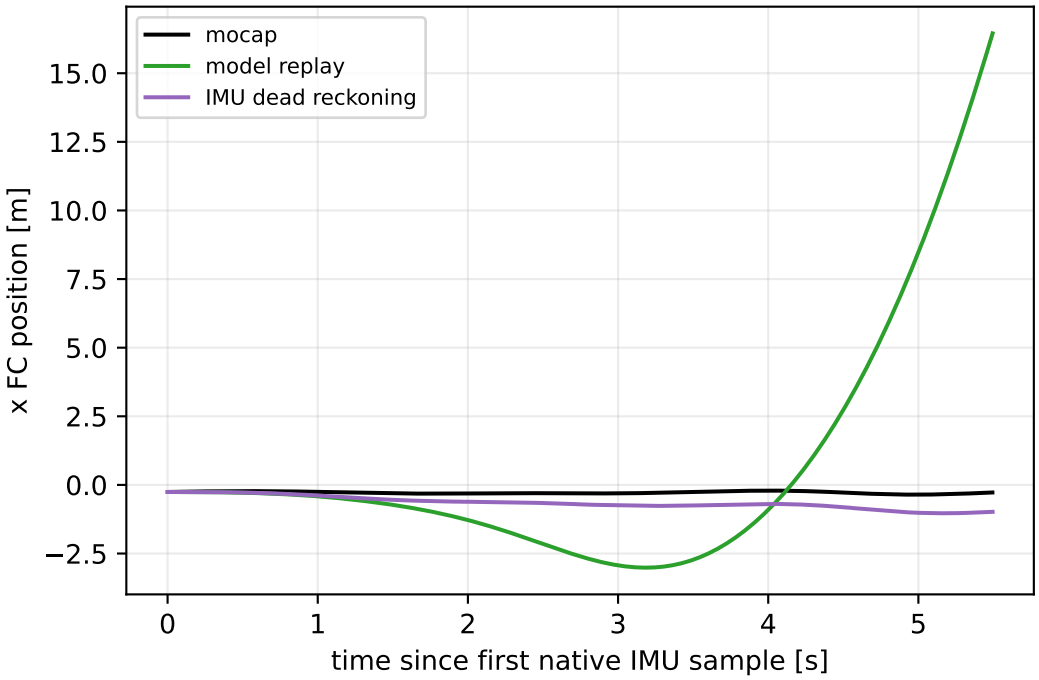

failure2 — SO(3) attitude errors at original mocap timestamps
GA: RMS 1.774°, final 1.997°
GA + magnetic output: RMS 16.910°, final 1.997°
model replay: RMS 45.934°, final 84.503°
geodesic attitude error [deg]
0
20
40
60
80
0
1
2
3
4
5
time since first native IMU sample [s]

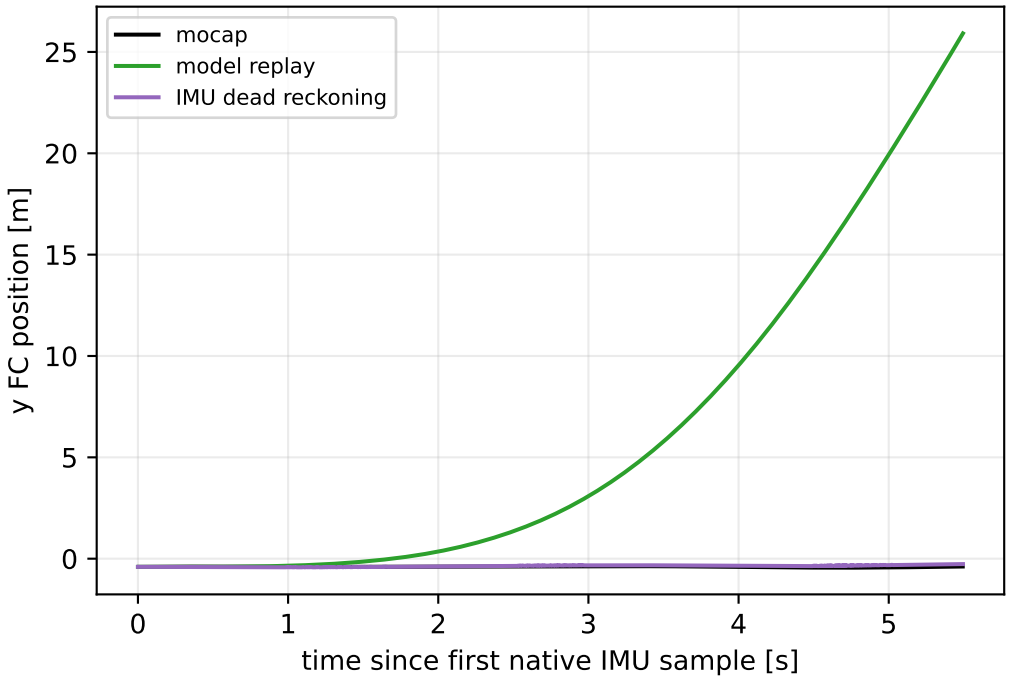

failure2 — GA: local attitude mean and ±2σ uncertainty
local Gaussian ±2σ
GA
mocap
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]

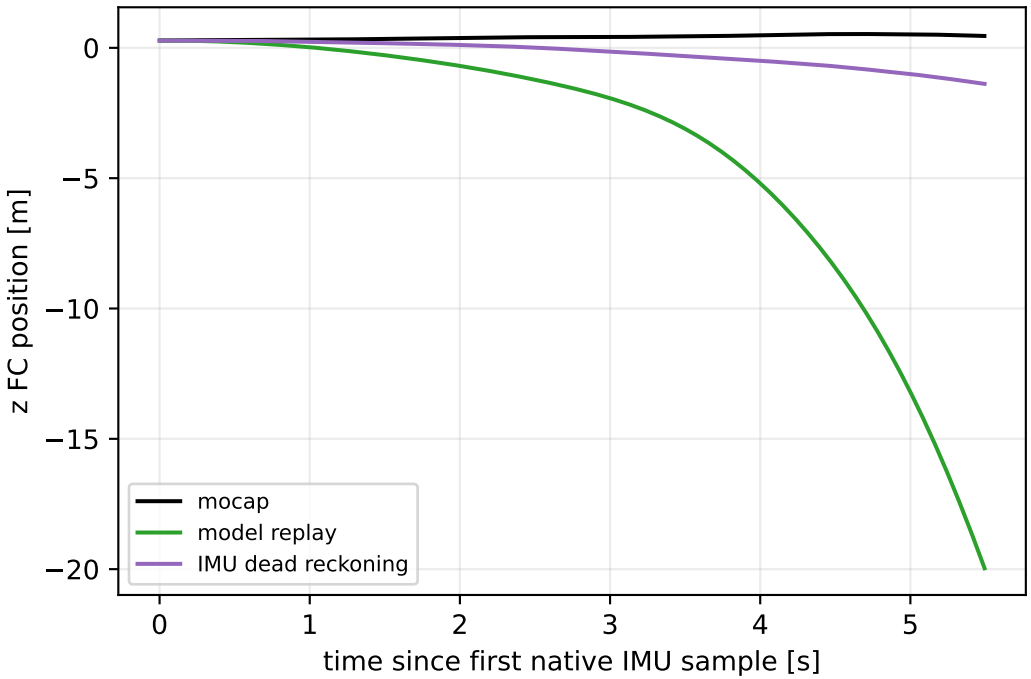

failure2 — GA + magnetic output: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
GA + magnetic output
mocap

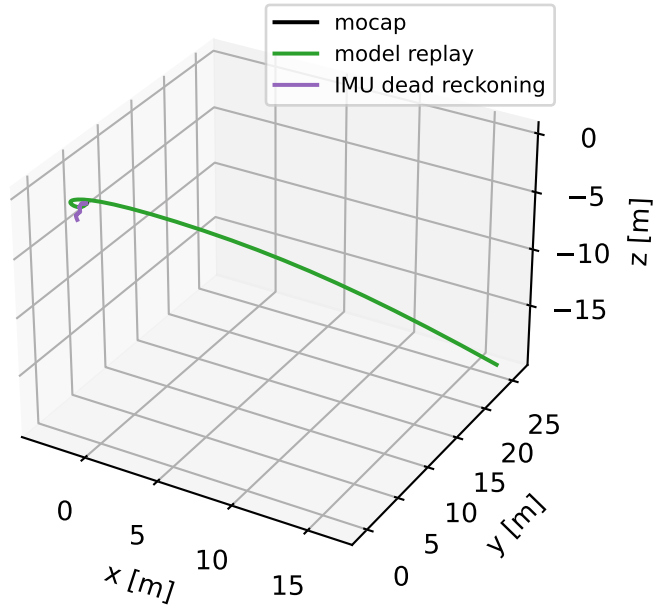

failure2 — model replay: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
local Gaussian ±2σ
model replay
mocap
time since first native IMU sample [s]
2000
0
−2000
1000
500
−500
−4000
1
2
3
4
5

failure2 — Position reconstruction (mean trajectories)



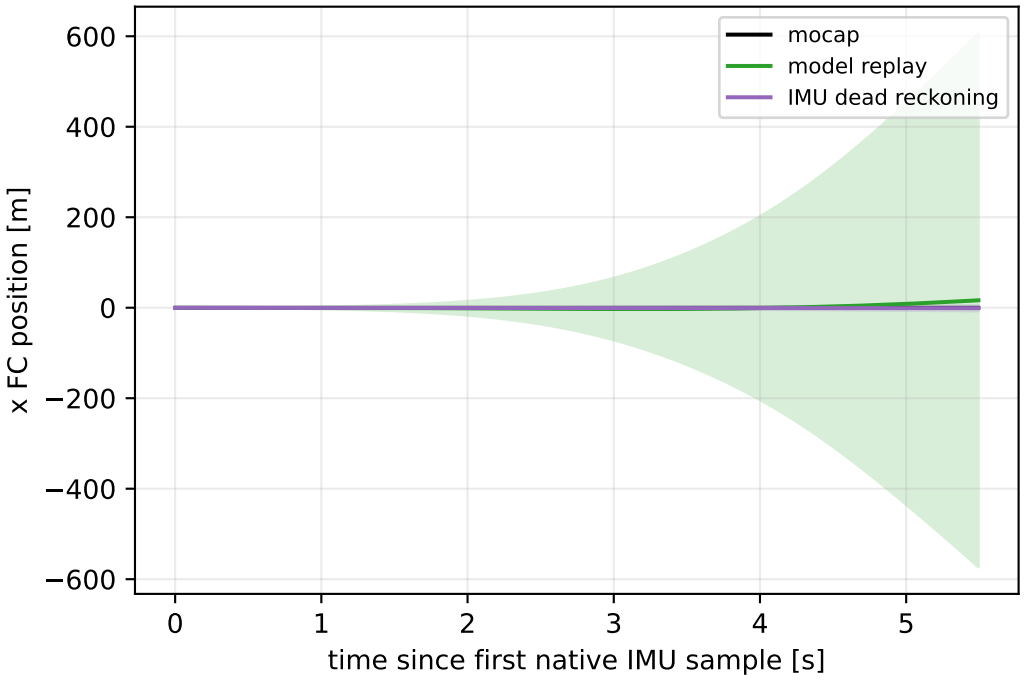


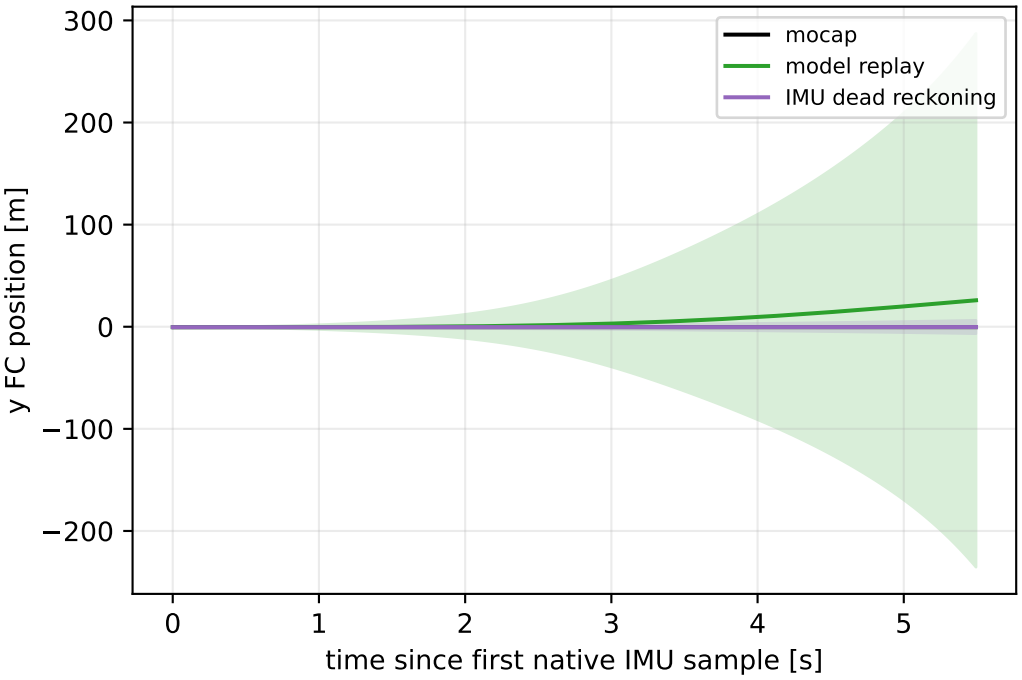


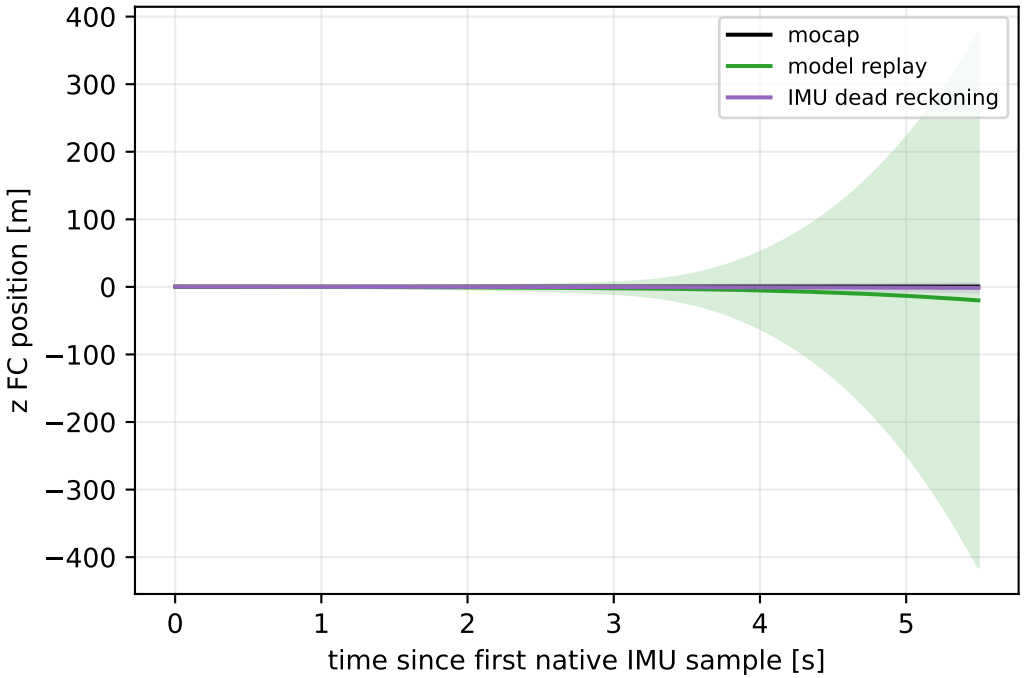


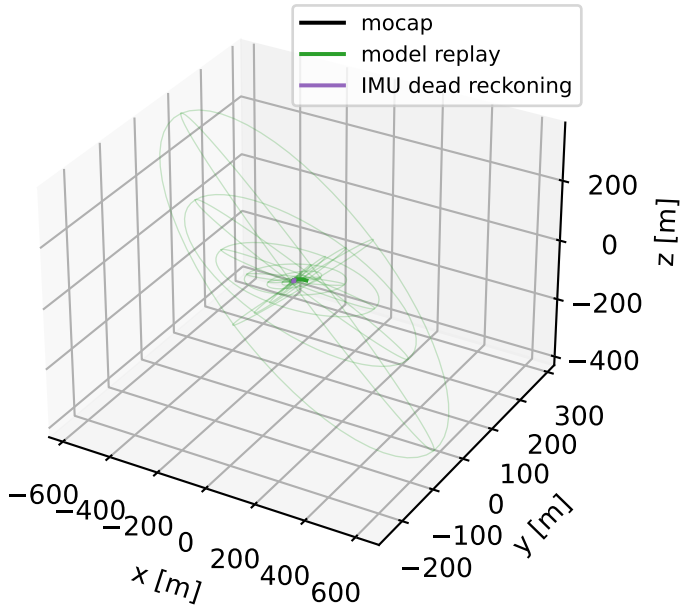
Pure physical replay and IMU integration are not reset to later mocap.

failure2 — Position reconstruction with full propagated ±2σ spread









FC origin; local Gaussian linearization. Ellipsoids: deterministic 2σ factor-SVD wireframes, not samples.